

RelieFoam: Rapid Prototyping of 2.5D Texture using Laser Cutter

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ABSTRACT

We propose RelieFoam, a rapid prototyping method that uses a laser cutter to produce a 2.5D textured surface on polystyrene foam. Conventional rapid prototyping methods using a laser cutter can quickly create 3D prototypes, but with the limitation that their surfaces can only be flat and smooth. Our method enables the rapid creation of objects with finely detailed 2.5D textured surfaces. By applying localized high-density energy from a laser cutter to low thermal conductivity polystyrene foam, a laser cutter can selectively remove only the specific parts of the surface of the polystyrene foam being targeted. We have built a computational model that calculates the laser parameters required to engrave the polystyrene foam in this way. Applying the laser parameters calculated by our model, we are able to implement applications with haptic textures and visual translucency and thus demonstrate the potential of our method.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Rapid prototyping, Texture, Laser processing, Foam

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1 INTRODUCTION

Rapid prototyping has become an important process for facilitating the quick embodiment and testing of concepts. Laser processing can be conducted at a higher speed than the general 3D printing method, and it has been widely adopted for quickly prototyping rough 3D-shaped objects. For example, research has been conducted on creating low-fidelity 3D prototypes by applying an attenuated laser beam to an acrylic board, heating it to be compliant and bent by gravity [6, 7]. Similar research has used an attenuated laser beam to weld acrylic boards together to create low-fidelity 3D prototypes [12]. In addition, several methods have been proposed

for the creation of 3D objects based on the assembly of boards laser-cut to 2D patterns which have been output by a software system [1–4]. These approaches allow the desired model to be constructed in a relatively simple operation that involves processing flat boards into a designed shape and assembling them into a final 3D object, either automatically or manually. While offering a very simple way of creating 3D objects, the surfaces of the prototypes produced using these methods are generally flat and smooth.

On the other hand, several prototyping methods using 3D printers have been proposed for manufacturing objects with customized textured surfaces [5, 8, 10, 11]. Fused deposition modeling (FDM) 3D printers can create uneven textures by controlling the printing path and the amount of material extruded during printing. However, 3D printing to create high-definition textures in this way is time-consuming.

Therefore, we aim to realize rapid prototyping of 3D objects featuring arbitrary textures and propose a method for creating high-definition 2.5D textured surfaces using laser processing. By employing polystyrene foam boards and applying appropriate localized heat with a laser cutter, we can selectively remove precise areas to a specific depth from the surfaces of the boards to form fine 2.5D textured surfaces.

2 PROPOSED METHOD

Our method uses a laser cutter to create 2.5D textured surfaces by selectively removing precise areas of polystyrene foam to a specific depth. A laser cutter can apply high-density energy locally and selectively and, when applied to polystyrene foam with its low heat transfer coefficient, it can concentrate heat and volatilize only the specific part of the surface targeted for removal, thus creating a fine 2.5D textured surface. However, if the laser output is set too high, the excess energy melts parts of the polystyrene foam that should remain, and if the laser output is not high enough, the targeted parts of the polystyrene cannot be removed properly. To determine the optimal laser parameters for our method, we investigated the depth of the penetration into the polystyrene foam achieved by the laser cutter under several conditions, and we built a simple model to calculate the laser parameters required to precisely remove targeted areas of the polystyrene foam.

We used a laser cutter (PLS6.150D, Universal Laser Systems, Inc.) to make cuts into a 5 mm thick sheet of polystyrene foam board (density ρ : 57.56 kg/m³, thermal capacity C_p : 1300 J/kg·K) by setting the following laser parameters: power p (1–4%), cutting speed s (1–6%) and pulses per inch (PPI: 500). The results (Figure 1) showed that when the laser power p was high, or the cutting speed s was slow, the polystyrene foam was cut deeper, but if the output energy was insufficient, it was observed that the sheet was only cut out halfway. Based on these results, we modeled the phenomenon

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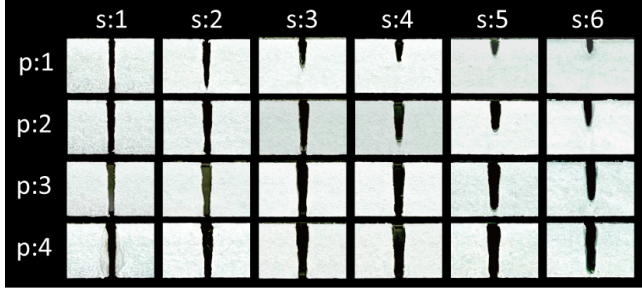


Figure 1: Side views of the depth cut by the laser into the polystyrene foam at different settings (s: Speed [%], p: Power [%]).

to clarify the optimum setting of the laser cutter for engraving textured surfaces on the polystyrene foam.

The heat energy Q_{laser} provided by the laser cutter per second is given by the following formula, where the maximum output of the laser cutter is P ($= 120$ W) and the laser power parameter p (%).

$$Q_{laser} = P \cdot p / 100 \quad (1)$$

Depending on the wavelength of the laser and the color of the material, some of the energy output from the laser cutter is reflected and wasted, so if the loss coefficient is η , the energy used to cut the polystyrene foam Q_{cut} can be expressed as follows:

$$Q_{cut} = \eta \cdot Q_{laser} \quad (2)$$

Meanwhile, the energy required per second to engrave the polystyrene foam Q_{cut} is given by (3) wherein ρ is the density of the polystyrene foam, C_p is the thermal capacity, T_1 is the thermal decomposition temperature of the polystyrene ($=364^\circ\text{C}$ [9]), T_0 is the ambient temperature ($=20^\circ\text{C}$), V_1 is the volume engraved in 1 second.

$$Q_{cut} = V_1 \cdot \rho \cdot C_p \cdot (T_1 - T_0) \quad (3)$$

For simplicity, the volume V_1 can be assumed as (4) if the depth engraved by the laser is d , the width engraved by the laser is w , the maximum plot speed of the laser cutter is S ($= 1270$ mm/sec in vector scan mode), and the speed parameter of the laser is s (%).

$$V_1 = d \cdot w \cdot (S \cdot s / 100) \quad (4)$$

The cutting depth d and cutting width w obtained in the above test cuts are substituted into the formulas (1) to (4) to calculate the loss coefficient η , and the average value of η is 0.94. Selecting the laser plotting speed s and laser power p based on the value of η allows the calculation of the appropriate engraving depth d of the laser into the polystyrene foam.

Based on the calculation model, we tested the processing of two-dimensional images on a 5 mm thick polystyrene foam board. We created an image of 1.5×1.5 cm squares filled in with black and laser processed it on the polystyrene foam board in raster scanning mode while changing the laser power parameter p (p : 13, 20, 26, 40, 53, 66%) and retaining a constant speed parameter (s : 100%). In raster scanning mode, a line was drawn at 500 PPI, and 90 lines per inch (i.e., 1 line width = 0.28 mm) were plotted at a maximum speed of 2540 mm/sec.

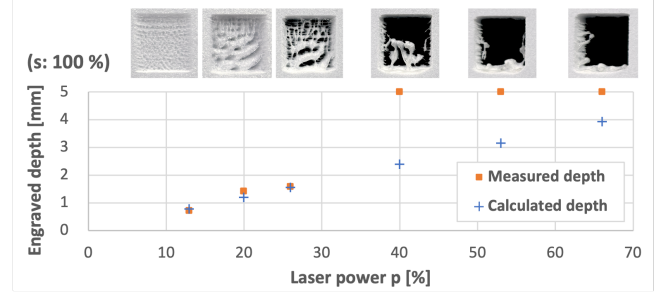


Figure 2: Measured depths compared to calculated depths for each laser power.

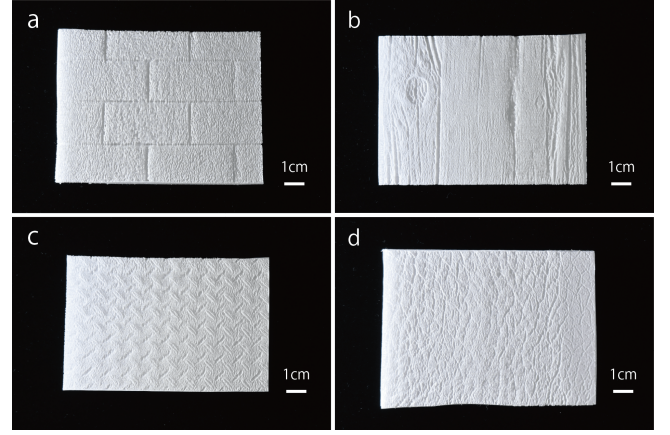


Figure 3: Example of haptic texture generation. (a) Tile (b) Wood grain (c) Tread plate (d) Leather.

The results are shown in Figure 2. When the engraving depth d is less than 2 mm (p : 13, 20, 26), the measured depth is almost the same as the calculated depth. However, as the value of p increases, unintended parts of the polystyrene seem to melt, allowing more light to pass through. When p is 40 or more, most of the processed area is melted. These results indicate that the heat supplied by the laser is diffused, and diffused energy has the effect of melting and shrinking the polystyrene foam rather than of volatilizing it. For this reason, we concluded that either the engraving depth must be less than 2 mm from the top surface, or the calculation model must include the heat diffusion factor to adjust the laser parameters.

3 APPLICATION EXAMPLES

Based on the results described in the previous section, we implemented several applications featuring haptic textures and visual effects using light transmission.

Haptic Textures. The proposed method enables the rapid prototyping of various textured surfaces. Detailed textures can be simply produced by sending any 2D image to the laser cutter and processing it using the laser-cutter parameters calculated in the way described in the previous section (Figure 3).

Invisible Object Tagging. The proposed method enables precise control of the light transmittance of the finished material by



Figure 4: Example of an invisible QR code which appears only when illuminated from behind.

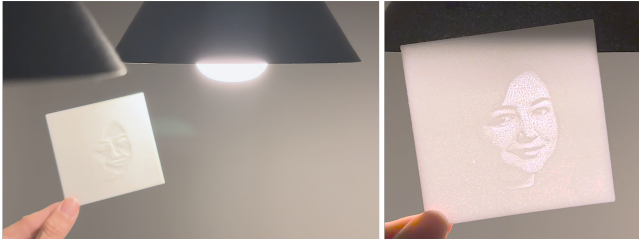


Figure 5: Example of a lithophane: the woman's face appears only when illuminated from behind.

partially engraving and thinning the polystyrene foam board. Consequently, when light is shone from behind the board, certain areas become translucent while others remain opaque, revealing a pattern. This can be applied, for example, to invisible object tagging, where tags appear only when illuminated from behind. Figure 4 shows an example of a scannable QR code that becomes visible when backlit.

Lithophane. Lithophanes are artworks where detailed images become visible only when a plate engraved with a certain pattern is illuminated from behind. The proposed method can quickly produce boards engraved with arbitrary patterns to rapidly create such lithophanes. Figure 5 demonstrates an example of a lithophane realized using our method. The surface of the board is engraved with a woman's face, and when illuminated from behind, a detailed facial image appears.

4 CONCLUSION

We have proposed RelieFoam, a rapid prototyping method for creating 2.5D textured surfaces using laser processing. By applying localized heat to a polystyrene foam board with a laser cutter, we can selectively remove precise parts of the surface of the board to form a fine 2.5D textured surface. We believe that this study contributes to a broadening of the range of prototype formable objects that can be created using laser processing.

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